KIEU STUDIO · GITHUB RADAR · EPISODE 35 · CHEAT SHEET

prometheus

A monitoring system: collect metrics, query with PromQL, alert.

github.com/prometheus/prometheus

+322	67,528	Apache-2.0
stars today · Trending lists no weekly or monthly figure	total stars · GitHub API	license · created Nov 24, 2012

Prometheus (Apache-2.0, a CNCF project) is a systems and service monitoring system: it collects metrics from targets at regular intervals, stores them as multi-dimensional time series, queries them with PromQL, draws graphs and fires alerts. Each single server is autonomous; federation is available.

How it works in six steps

1. **Find the targets** — discovery or static config
2. **Pull metrics on a schedule** — pull model over HTTP
3. **Store time series** — each server is autonomous
4. **Query with PromQL** — across many dimensions
5. **Draw graphs** — web UI, dashboards
6. **Alert** — when a condition occurs

Quick start

```
# Try it with Docker
docker run --name prometheus -d -p 127.0.0.1:9090:9090 prom/prometheus

# Run with a config file
prometheus --config.file=prometheus.yml

# Build it yourself
git clone https://github.com/prometheus/prometheus.git
cd prometheus
make build

# Open the web UI
http://localhost:9090

# Needs: Docker, or a compiled binary; building from source needs Go, NodeJS and npm
```

Key commands / features

- `prometheus --config.file=...` — run the server with a config file
- `promtool` — companion binary, built with prometheus
- `make build` — build, including the web UI
- `make test` — run the tests
- `make format` — format the source

Growth



Two real points: the start of the window and today, axis from zero. The gain is as reported by GitHub Trending; we did not rebuild the star history.

Why this repo exists

- A system running for real needs to know how healthy it is: how much load, how many errors, where it is slow.
- Collect metrics on a schedule, query across dimensions with PromQL, and alert when a condition occurs.

What to use it for

- Monitor load, errors and latency of services
- Query metrics across dimensions with PromQL
- Set alerts for the conditions you worry about
- Find targets automatically with service discovery

Think twice before you use it

- Pull model: the server must be able to reach the targets it monitors.
- Pushing metrics (short jobs) needs an intermediate gateway.
- No distributed storage: scale out with federation.
- The README's Docker example binds port 9090 to 127.0.0.1: keep it that way when you try it.

Sources: the repo's README, https://github.com/prometheus/prometheus, https://prometheus.io. Stars: GitHub API; gains: GitHub Trending (as reported by Trending), snapshot Nov 8, 2026; stars measure attention, not quality: re-check before relying on them. Summarized in Kieu Studio's own words; commands copied from the README; no third-party logos or screenshots.
Disclosure: drafted with AI assistance; the voice and presenter in the videos are AI-generated. Not investment advice, and not endorsed by Kieu Studio. **kieu.app**